ShuttleAide : IIUM Bus Schedule & Tracking Assistance

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***Abstract*—Efficient campus transportation is critical for universities like IIUM Gombak, where shuttle buses are a vital eco-friendly commuting option. However, students face challenges such as scattered bus schedules, lack of real-time tracking, and inefficient communication of updates. ShuttleAide addresses these issues by integrating IoT and GPS technologies to provide a user-friendly mobile application. The project followed the Rapid Application Development (RAD) methodology, emphasizing iterative design and refinement. The system includes real-time bus tracking, schedule access, and announcements. The IoT-based GPS system, built with an ESP-WROOM-32 microcontroller and Neo-6M GPS module, transmits location data to Firebase for seamless integration with the app. The implementation successfully meets its objectives, with the UAT revealing that all features passed 100% except for real-time tracking and ETA, which experienced one failure each out of three tests. Overall, ShuttleAide provides an innovative and efficient solution to streamline campus transportation, enhancing user experience and reliability for the IIUM Gombak community.**

Keywords: Campus Transportation, IoT, GPS, Real-Time Bus Tracking, Database, Flutter, Mobile App

I. Introduction

Efficient campus transport systems are crucial for ensuring the smooth functioning of educational institutions [1], such as International Islamic University Malaysia (IIUM) Gombak. Shuttle buses play a vital role in facilitating eco-friendly in-campus transportation, providing students and university residents with a reliable means of travel between various locations. Although the shuttle bus service is a vital mode of transportation, it currently faces significant challenges.

Accessing the current bus schedule for IIUM Gombak's campus is a major challenge faced by students, as schedules are scattered across the university website and messaging platforms like Telegram and WhatsApp. Inefficient communication of updates and announcements exacerbates the issue, resulting in commuters being inadequately informed about service updates. Additionally, students face prolonged waiting times at bus stops without the ability to track shuttle bus locations, leading to uncertainty and dissatisfaction.These challenges highlight the need for an efficient and centralized solution to improve the usage of shuttle bus service.

ShuttleAide is a mobile application developed to address these issues by combining Global Positioning System (GPS) and Internet of Things (IoT) technology to provide a streamlined and user-friendly experience for campus transportation. The application offers real-time bus tracking, centralized access to updated schedules, and announcements for service updates. This solution aims to enhance the efficiency, reliability, and overall user experience of the shuttle bus service.

The objectives of the ShuttleAide project include gathering functional and non-functional requirements to design a mobile application that offers real-time bus tracking, developing a cost-effective IoT-based GPS tracking embedded system, and evaluating the usability and functionality of the final application. In addition to addressing the immediate needs of campus commuters, ShuttleAide aligns with IIUM's broader goals of creating an eco-friendly and efficient campus environment.

Moving forward, this paper comprises Related Works, which explores existing solutions and technologies; Methodology, which outlines the development approach and processes; and Results and Discussion, which highlights the system's implementation, testing outcomes and considerations. Together, these chapters provide a comprehensive overview of the ShuttleAide project and its contribution to addressing IIUM Gombak's transportation challenges.

II. Related Works

A. Existing Systems

An in-depth analysis was conducted on five mobile applications relevant to the ShuttleAide project: Moovit, MyRapid PULSE, ViaBus, Grab, and inDrive. Moovit, launched in 2012, is a free application designed to help users navigate cities efficiently using various modes of transportation [2]. Similarly, MyRapid PULSE, introduced by Prasarana in 2020, assists public transport commuters in the Klang Valley, Penang, and Kuantan [3]. ViaBus, developed by a Thai tech startup in 2018, focuses on real-time public transport tracking and navigation [4]. Grab, Southeast Asia's leading super app, offers a range of services such as ride-hailing, food delivery, and bill payment, with e-hailing being its primary feature [5]. Lastly, inDrive, formerly known as inDriver, is an international ride-hailing service that ranked as the second most downloaded ride-hailing app globally in 2022 and 2023 [6].

The advantages and disadvantages of each system are listed in Table I below.

TABLE I. EXISTING SYSTEMS REVIEW

System	Advantages	Disadvantages
All	Interactive map display	Interaction can be laggy at times
	Usage of slide-panel for easy access to hidden information	
Moovit	Display full route of specific bus	Intrusive advertisements pop up
	Schedules & Real-time tracking	Real time location tracking & arrival updates require premium subscription
	ETA Display	
	Ability to favorite specific lines, or destinations	
MyRapid PULSE	Schedules & Real-time tracking	Significantly unreliable bus arrival time on certain occasions
	Real-Time Service Alerts	
	ETA Display	Design prioritizes functionality over aesthetics
	Allows users to filter what to display on the map	
ViaBus	Real-time station tracking (Not bus tracking)	Live arrival times and alerts require premium subscription
	Ability to view traffic lines	Does not have real-time bus location tracking
Grab	Real-time vehicle tracking	Does not provide freedom in pinning specific locations
	ETA display	Nearest location feature dysfunction at times
inDrive	Real-time vehicle tracking	Heavily reliant on strong internet connections
	ETA display	Too much display on the screen

B. *Research Paper*

One study proposed the solution of real-time tracking of vehicles with the usage of IoT modules such as Arduino Uno R3, SIM800L, and NEO 6M GPS, and it is web-based [7]. The study found that the integration of the system enables comprehensive monitoring, and features like historical location tracking, speed monitoring, and geo-fencing, enhancing efficiency, and safety.

Another study proposed a solution that is designed with the usage of ESP32 with a Wi-Fi built in module, a GPS module, and the mobile app [8]. The study shows that with the integration of the system in a mobile app, it ensures seamless information access.

A study proposed a solution of vehicle tracking systems that uses GPS and Global System for Mobile Communications (GSM) technology for a cost-effective system [9]. The proposed solution tackles both issues which is the information accessibility and availability, ensuring the information is accessible and available via Short Message Service (SMS) to the users mobile phone. However, the GSM might cause an issue with the connectivity delays caused by its networks.

Other studies proposed a solution that utilizes Artificial Intelligence (AI) for vehicle tracking and monitoring [10]. The system offers high accuracy of location, supported with A-GPS technology. Although the system does offer accurate location, the technology proposed costs much more, and the solution cannot send data to the cloud which limits real-time data sharing.

In conclusion, a mobile app that integrates with IoT technology are essential tools for students to access bus schedules, and have real-time tracking of the campus shuttle bus. The research papers reviewed proposed different approaches to developing a real-time bus tracking solution, including mobile-based application, web-based systems, and various types of GPS and IoT. These solutions aim to help those students that faced the prolonged issues in riding the shuttle bus.

III. METHODOLOGY

The development of ShuttleAide followed a structured and iterative approach to ensure the system met the requirements of IIUM Gombak's campus community. The methodology employs the Rapid Application Development (RAD) approach, whose flexibility reduces planning time while increasing the focus on prototyping [11], facilitating quick iterations and ensuring continuous feedback incorporation for refinement.

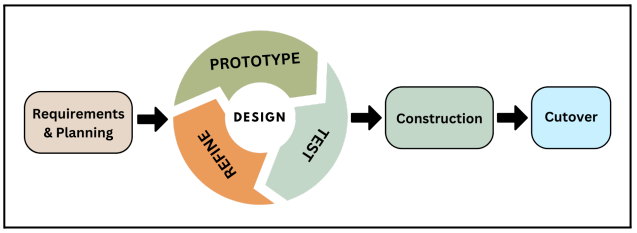


Fig. 1. Rapid Application Development Approach

From Fig. 1, The RAD process is structured into four distinct phases: Requirements & Planning, Design, Construction, and Cutover. Each of the phases are structured as below :-

A. *Requirements & Planning*

The first phase involves collecting requirements to define the system's essential features. This is made concrete through methods such as reviewing existing systems, research papers, and gathering feedback through a questionnaire to potential stakeholders. These insights

clarify stakeholders' needs and expectations, guiding product development and establishing precise system requirements.

The use case in Fig. 2 below illustrates the context and functionalities of the ShuttleAide system.

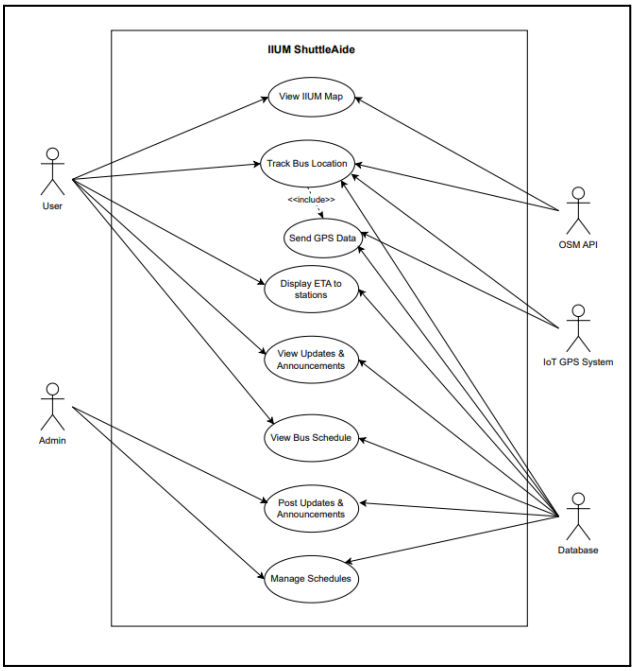


Fig. 2. Use Case Diagram

The use case diagram includes five actors: users and admin as primary actors, and OpenStreetMap Application Programming Interface (OSM API), IoT GPS System, and the database as secondary actors. Users can view the IIUM map, track buses, check Estimated Time Arrival (ETA), access schedules, and see updates or announcements. Admin manages schedules and posts updates, with data stored in the database. The system integrates real-time data from the IoT GPS and map data from OSM API for efficient functionality.

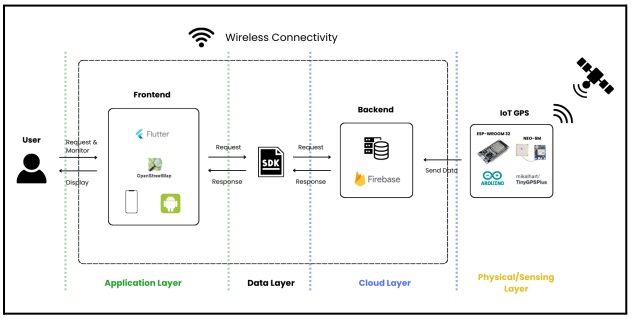


Fig. 3. System Architecture Diagram

The system architecture is designed to provide an overview of how elements are connected to each other in the ShuttleAide project ecosystem. Based on Figure 3, the system architecture integrates multiple components, enabling seamless interaction between IoT devices, the Firebase database, and the ShuttleAide mobile application.

The system begins with the Physical/Sensing Layer, where IoT devices like Arduino, GPS Neo-6M, and ESP32, using TinyGPSPlus, collect GPS signals and wirelessly transmit data to the backend. In the Cloud Layer, the backend processes this data, interacting with the Data Layer, where Cloud Firestore stores and retrieves GPS data for real-time use. Finally, the Application Layer presents the processed data via a Flutter-based mobile app with OSM integration, allowing users to view real-time location updates. Wireless connectivity ensures smooth interaction across all layers.

B. Design (Iterative)

The design phase of the ShuttleAide project focused on creating and refining the user interface, developing prototypes, and implementing both the IoT-based GPS tracking system and backend functionality.

The user interfaces for ShuttleAide were initially designed using Figma, guided by the functional and non-functional requirements identified during the requirements specification stage. Key prototypes included the Home Page, ETA Page, Schedule Page, and Announcement Updates Page. These designs formed the foundation for translating visual concepts into functional components of the mobile application.

Prototypes were developed iteratively, beginning with the frontend. The frontend, built using the Flutter framework, transformed the Figma prototypes into Dart code. Emphasis was placed on delivering a seamless user experience (UX) and ensuring interactive features such as real-time bus tracking and dynamic schedules were functional and user-friendly.

The IoT GPS tracking system utilized an ESP-WROOM-32 microcontroller paired with a Neo-6M GPS module. This hardware configuration allowed the collection and transmission of real-time GPS data, which was critical for tracking shuttle bus locations. The specific wiring and setup details of the GPS system are illustrated in Fig. 4 and Table II.

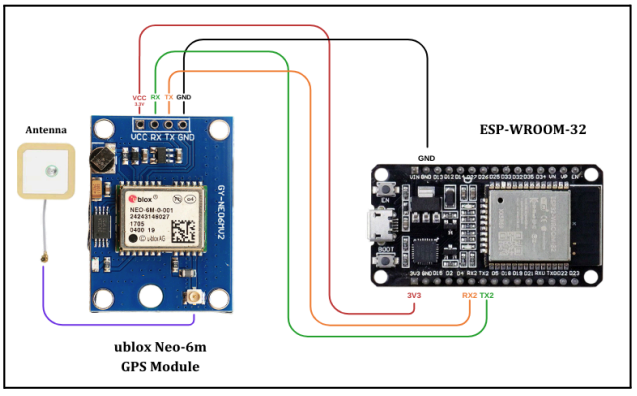


Fig. 4. IoT GPS System Circuit Diagram

TABLE II. GPS WIRING CONNECTION

	ESPWROOM-32	Neo-6M GPS
Connection	3V3	VCC
	GND	GND
	RX2	TX
	TX2	RX

The backend implementation relied on Firebase for real-time data handling. The Realtime Database stored and updated bus location data, including latitude and longitude coordinates sent from the GPS tracker. Firestore was employed to manage static data, such as bus schedules and announcements, ensuring that users could access updated information effortlessly.

Through iterative design and development, this phase ensured all components worked cohesively, laying the groundwork for a fully functional and integrated system.

C. *Construction*

The construction phase brought together all components of the ShuttleAide system, including the frontend, IoT-based GPS tracking, backend, and additional features, ensuring a cohesive and fully functional application.

The interactive map, built with OpenStreetMap using the flutter_map package, serves as the core of the application, allowing users to view shuttle bus locations and campus bus stops.

For real-time GPS location display, the Neo-6M GPS module, connected to the ESP-WROOM-32 microcontroller, transmits latitude and longitude data to the Firebase Realtime Database using the TinyGPSPlus and Firebase-ESP-Client libraries. This data is then retrieved by the mobile application to dynamically update the moving bus markers on the map, enabling real-time tracking.

Schedule and announcement management is handled through Firestore, where administrators manually input data via the Firebase console. The mobile application fetches and displays this data on the schedule and updates pages, ensuring users have access to accurate and up-to-date information.

Additional features were implemented to enhance the user experience. A tweening algorithm ensures smooth movements of the bus markers, while ease-in animations were added for seamless map transitions and sliding panel interactions. Popup markers allow users to tap on bus stops to view descriptions and images for better navigation. Lastly, a follow-bus feature lets users select a bus, causing the map view to automatically follow its movement.

This phase ensured all system components were integrated effectively, delivering a streamlined and user-centric transportation solution.

D. *Cutover*

The Cutover phase is where the ShuttleAide system is prepared for system deployment. User acceptance testing (UAT) and other final system testing are carried out to make sure the application satisfies all requirements and functions as intended.

This structured methodology ensures that ShuttleAide is not only a technically robust application but also a user-centric solution, addressing the transportation challenges faced by the IIUM Gombak community.

IV. RESULT

According to the survey, many potential users are eager to support the mobile application, ShuttleAide.

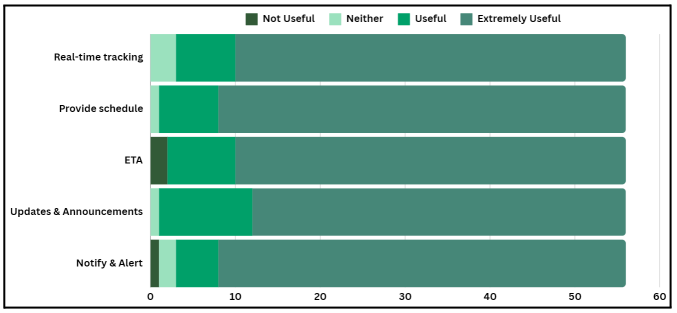


Fig. 5. Functional Requirements

Based on Fig. 5, the questionnaire responses highlight key priorities for the system. For functional requirements, 82.1% of respondents emphasized the need for real-time bus tracking, while an equal 82.1% prioritized ETA information for stations. Additionally, 85.7% requested a centralized schedule viewing feature, and 78.6% wanted announcements and updates from administrators.

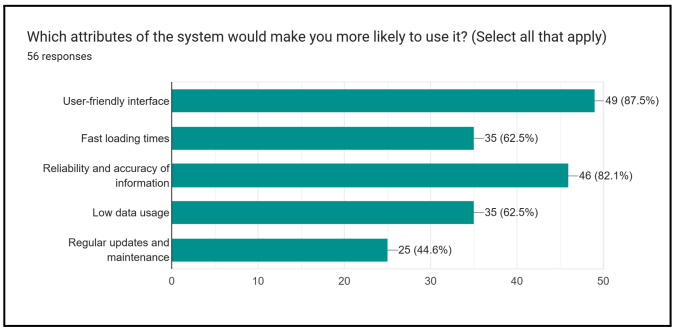


Fig. 6. Non-Functional Requirements

As for the non-functional requirements, Fig. 6 shows that 87.5% of respondents stressed the importance of a user-friendly interface, 82.1% highlighted the need for reliable and accurate information, and 62.5% called for fast-loading application performance. These insights form a strong foundation for system design.

A. *System Interface*

The implemented interfaces of the ShuttleAide mobile application can be seen in the figures Fig. 7. until Fig. 12..

Fig. 7. shows the main page of ShuttleAide. Fig. 8. shows the sliding panel that will contain the bus information, including the ETA. Fig. 9. and Fig. 10. shows the result of the bus tracking feature where Fig. 9. shows the bus location at time 17:21, and Fig. 10. showing the corresponding location at time 17:22. The two figures illustrate how, upon retrieving new location data, the bus tracking feature updates the bus's location on the app display. Lastly, Fig. 11. and Fig. 12. shows the schedule and announcement page.

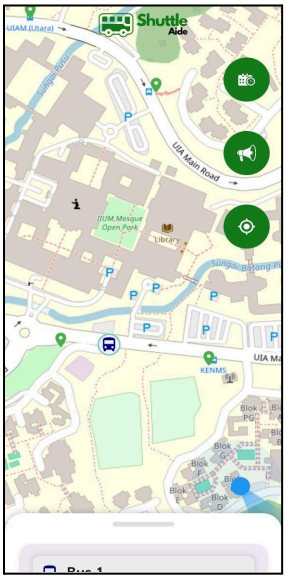


Fig. 7. Main Page

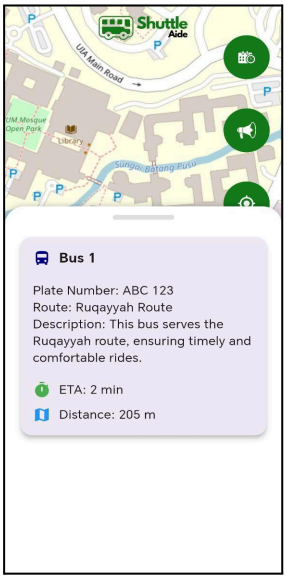


Fig. 8. ETA

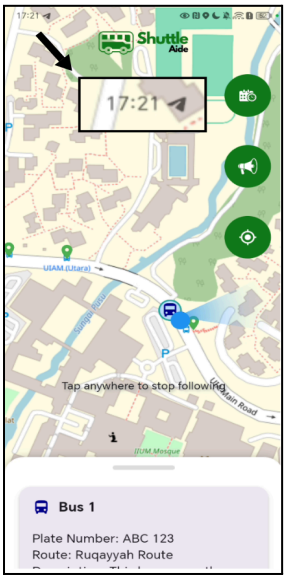


Fig. 9. Bus Tracking T1



Fig. 10. Bus Tracking T2

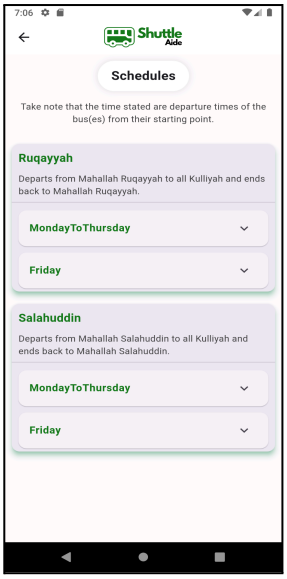


Fig. 11. Schedule Page

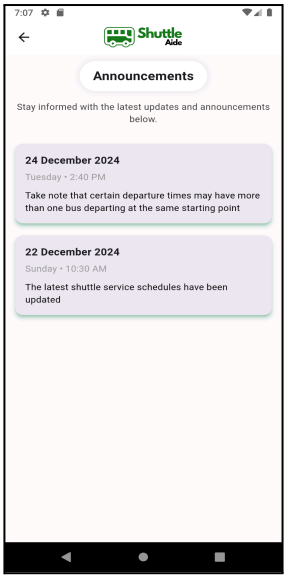


Fig. 12. Announcement Page

B. *Testing*

Testing is a critical phase in the ShuttleAide project to ensure its components function correctly, integrate seamlessly, and meet user expectations [12]. Three tests of User Acceptance Testing (UAT) are done to evaluate the system. Fig. 13 until Fig. 19 displays the percentage of results of the tests of each feature in the system.

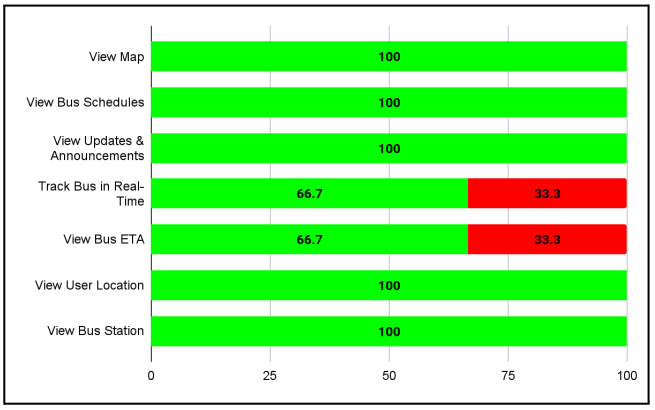


Fig. 13 UAT Results

Out of the test results, only the real-time tracking and ETA features failed once out of the three times each. On the other hand, other features passed flawlessly for all the three tests conducted.

V. DISCUSSION

ShuttleAide is a mobile application designed to enhance campus transportation at IIUM Gombak. It provides real-time bus tracking, centralized access to updated schedules, and announcements for service updates. The system integrates IoT and GPS technology to deliver a user-friendly and efficient solution for students, staff, and visitors navigating the campus shuttle service.

A. *Limitations*

Despite its success, the ShuttleAide project encountered several limitations. The current antenna of the GPS module struggles to acquire signals in areas without a clear view of the sky. Additionally, the ESP-WROOM-32 microcontroller relies on WiFi for transmitting GPS data to Firebase, making the system dependent on stable internet connectivity. Another limitation involves the power source for the IoT device, as no dedicated solution has been implemented; this could affect the reliability of the system in a real-world deployment. Lastly, the free Firebase plan used for this project restricts daily reads and writes with the Realtime Database only allowing 100 simultaneous connections at one time[13], which may not scale well with a large user base.

B. *Recommendations for Future Work*

Several enhancements are recommended to improve the system. Incorporating an external GPS antenna would strengthen signal reception, ensuring reliable performance in challenging environments. To eliminate WiFi dependency, GSM modules or Subscriber Identification Module (SIM) card-based cellular WiFi systems could enable standalone data transmission. Developing a dedicated and sustainable power solution, such as rechargeable batteries or integration with the bus's electrical system would enhance the system's long-term reliability. For database scalability, transitioning to a more robust plan or exploring alternative database solutions could support broader deployment.

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